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Which Electoral Districts Get More Money

The Japanese politics literature exhibits a puzzle. Time and again, we are told that Liberal Democratic Party (LDP) politicians use geographically targeted spending to cultivate loyal groups of supporters, who can be relied on to turn out and vote for them on election day.[1] Some studies go further and argue that it is these politicians' access to geographically targetable resources, as well as to other areas of government policy that can be parlayed into benefits for nationally organized groups of voters, that constitutes the most compelling explanation for the LDP's ability to vanquish its opponents, election after election, regardless of trends in its popularity, the state of the economy, or the electoral system. Yet whenever scholars have subjected this claim to empirical scrutiny, by collecting and analyzing data on votes and money, they have come up short. By this, I mean they have not furnished compelling evidence that spending is, in fact, channeled toward the party's supporters.[2] The lack of evidence for a claim that occupies a central place in scholarship on the party is curious and demands further investigation. If it is the product of restricting one's gaze to a handful of bad apples, who distort government resources in ways that might be troubling but are not generalizable to the broader pool of LDP politicians, then we should be looking elsewhere for explanations of the party's longevity at the reigns of power.

In this book, I offer a theory for how incumbents operating under a certain configuration of political institutions can use government resources to stay in power. The theory holds that whenever incumbents can discern

[1] Studies that make this claim include Reed (2021); Christensen and Selway (2017); McMichael (2018); Naoi (2015); Reed, Scheiner and Thies (2012); Saito (2010); Krauss and Pekkanen (2010); Pempel (2010); Estevez-Abe (2008); Hirano (2006); Scheiner (2006); Bouissou (1999); Tani (1998); Ramseyer and Rosenbluth (1993); Sone and Kanazashi (1989); Curtis (1971); and Thayer (1969).

[2] This is also a point made by McMichael (2018).

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their electoral support at a disaggregated unit within their electoral district and influence allocations of central government resources to those same units, they will have the tools to conduct what I call "group-based clientelism" (GBC). By this, I mean they will have the ability to *tie* the amount of government resources each unit (group) receives to its voting behavior. Moreover, when there is a dominant party, I explain that politicians in that party can elicit the most support by pitting groups against each other in a *competition* for resources. By promising the most resources to the group that ranks first in electoral loyalty, the second-most resources to the group that ranks second, and so on, and making sure that the difference in amounts received by the first and second place-getter are larger than the difference in amounts received by the second and third place-getter, and so on, politicians in the dominant party can maximize the number of votes won, conditional upon resources delivered. In this competition, a "tournament," the amount of money groups receive is a function of how their level of electoral support stacks up against that of other groups in the same district. The group ranking first in electoral loyalty receives the most money (the largest prize), with the amount of money groups receive declining in rank in a *convex* (rather than linear) fashion.

In Chapter 5, I explained that the institutional context in which the majority of LDP politicians found themselves in for the better part of the postwar period met the conditions under which they would have been able to implement tournaments. More specifically, these politicians competed in electoral districts consisting of groups of voters (municipalities), at which vote tallies were observable and government resources (national treasury disbursements [NTD]) deliverable. Beginning in the mid 1990s, I explained how the combination of electoral reform (1994), municipal mergers (2001–2005), and urbanization reduced the share of electoral districts in which tournaments are possible to approximately half of all districts. In Chapter 6, I examined whether LDP politicians use tournaments in tournament-possible districts. I explained how I gathered data on the voting behavior, resource allocations, and other fiscal and demographic characteristics of the universe of Japanese municipalities in existence from 1980 until 2014. Using regression specifications designed to minimize the influence of confounders, weigh up evidence for rival theories, and take advantage of distinctive features of the two electoral systems Japan used during this time, I presented evidence consistent with tournaments being used by many LDP politicians. My findings show that the amount of money Japanese municipalities received in the years following Lower House elections has been a function of how their support stacked up against that of their competitors in the tournament (other

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municipalities in the same district). Moreover, the amount of money municipalities received in the years following these elections declined with rank in exactly the (convex) manner predicted by the theory.

In contrast to prior work, then, I was able to unearth a powerful connection between money and votes, and unmistakable evidence that money flows to the party's core supporters. Putting it like this, however, risks misconstruing the rule under which resources were allocated. Resources are not *targeted* at core supporters (or municipalities with higher concentrations of core supporters) on account of the fact that these voters are core supporters (or are inhabited by core supporters) and the party is making a concerted effort to buy votes from core supporters. Rather, resources are used as a *reward* for the high level of support they received and are allocated without regard to the *type* of voters in each group. In this way, core supporters are given no special advantages over other types of voters. They can expect to receive disproportionately large government outlays only if they continue to return disproportionately high levels of electoral support. This was clear when I zeroed in on municipalities that exhibited the highest or second-highest level of support for their district's LDP winner(s) (Table 6.5). I found that when those municipalities slipped in rank, they suffered declines in size of their resource allocation.

In this chapter, I change the focus from money flows *within* electoral districts to money flows *across* electoral districts. As I explained in Chapter 4, a vast scholarly literature has been interested in the question of why voters in certain electoral districts receive larger allocations of (discretionary) government resources than voters in other districts, even after controlling for a host of other (nonpolitical) determinants of resource allocations.[3] Various theories have been offered. Most boil down to the idea that variation in resource allocations across districts is a function of the *type* of voters residing in each. Whenever resource allocations across electoral districts are found to be *positively* correlated with district-level vote shares for the ruling party, then the ruling party is presumed to be favoring core supporters (e.g., Cox and McCubbins 1986). Theoretical models offer cogent arguments for why it makes sense to target core supporters. Whenever resource allocations across districts are found to be *negatively* correlated with vote shares for the ruling party, meaning that districts where the party received less support got more money, then the

[3] Studies that have been interested in this question include Rickard (2018); Zarazaga (2016); Tavits (2009); Cox (2009); Stokes (2005); Dahlberg and Johansson (2002); Dixit and Londregan (1996); Lindbeck and Weibull (1987); and Cox and McCubbins (1986).

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ruling party is presumed to be favoring swing voters (e.g., Stokes 2005). A different set of formal theories provide the theoretical intuitions for this claim.

The tournament theory offers a different explanation for why resource allocations can vary so substantially across electoral districts, which moves scholarship in this area beyond the core-versus-swing dichotomy (Smith, Bueno de Mesquita and LaGatta, 2017; Smith and Bueno de Mesquita, 2012, 2019). To the extent that resource allocations vary across electoral districts, the theory holds that this will have little to do with the type of voter in each. Instead, it will be a function of the *ease* with which tournaments can be administered. Ease, Chapter 4 explains, is a function of the relative sizes of the groups of voters in each district. Why does relative size matter? To recap, tournaments work by creating an alternative dimension upon which voters can pivotal. When incumbents structure prizes so that they become exponentially larger at higher ranks, then provided that those prizes are sufficiently valuable, this encourages voters in municipalities with a chance of attaining a high rank to set aside whatever true feelings they may hold about the incumbent or her policies and turn out and support her in elections. As voters in all supportive municipalities come to this realization, turnout and electoral support increases, such that there is now little difference in support between the most supportive municipalities. This creates a situation where a very small number of votes in a supportive municipality can effect a change in rank, which can have big implications for a municipality's resource allocation. Critically, voters in supportive municipalities know that their votes will not matter to the *outcome* of the election but have been made to matter enormously for their municipality's *resource allocation*.

For a tournament to be an effective means of increasing support, however, voters have to be made *aware* that the incumbent plans to compare the performance of their municipality with that of other municipalities in the same district. But incumbents cannot make this explicit. As Chapter 4 explains, in a democracy politicians are supposed to vie for the support of voters, not make *voters* compete for *politicians*' time, attention, and resources. Because tournaments are antithetical to democracy (and indeed, threaten it), the incumbent's intention to rank groups based on electoral support and allocate resources on the basis of rank will all be implicit. This implicitness causes more problems when groups are asymmetrically sized than when groups are similarly sized. This is because there are different ways of comparing groups, and when groups are asymmetrically sized, those different ways of comparing groups can produce radically different results. Because different metrics lead to different results, and incumbents cannot easily *clarify* which metric they plan to

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use, voters in districts characterized by asymmetrically sized groups will have a great deal more trouble calibrating the degree of influence their vote holds over their municipality's ranking (and thus, its resource allocation). When voters have trouble calibrating the degree of influence their vote holds over their group's ranking (and hence, its resource allocation), they will be less inclined to turn out and support the incumbent.

As a result, the theory expects that incumbents in districts consisting of asymmetrically sized groups will have to offer *larger* prizes to convince the voters in their districts to use their votes to compete with other groups. This is Hypothesis 3. However, because incumbents will also prefer to minimize the amount of government money that has to be diverted to prizes, it is unlikely that they will deliver prizes large enough to totally offset the reduced propensity to turn out and vote for them among voters in these districts. The implication is that districts consisting of asymmetrically sized groups are likely to exhibit *lower* levels of electoral support, even as they receive *larger* prizes. This is Hypothesis 4. Looking across districts, then, the theory expects that resource allocations will be negatively correlated with electoral support. While resource allocations will be positively correlated *within* districts (due to the tournament), they will be negatively correlated *across* districts (due to variation in the ease with which tournaments can be administered). In other words, districts with less support for the LDP incumbent will receive more transfers.

Critically, this pattern will make it look like the LDP is *rewarding* districts for lower levels of support and *penalizing* districts for higher levels of support. Extant theories might attribute this to the party reasoning that, because core supporters are going to vote for it anyway, it would be prudent to direct geographically targeted resources toward swing voters. The tournament theory helps us see that this would be the *wrong* inference to draw about the determinants of allocations across districts, at least when politicians are using tournaments. The correct inference is that in districts in which groups vary greatly in size, voters require larger prizes to be drawn into a competition. This is not due to any intrinsic feature of those voters; rather, it is due to the relative sizes of the groups into which they are arranged. LDP politicians in these districts have to offer more resources than their counterparts in other districts. Because providing more resources is costly in terms of the politician's time and detracts from other policy goals, we can expect that politicians will settle for less support in these districts. But even this lower level of support will cost significantly more than a similarly low level of support in a district consisting of evenly sized groups.

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This macrolevel expectation of the tournament theory provides a new explanation for why prior studies of the political determinants of transfers in Japan did not furnish compelling evidence that within electoral districts, money flowed to core supporters. The fact that districts can vary so significantly in the amount of money needed to entice voters to compete with each other means that two otherwise-similar municipalities could be offering the *same* level of support but be receiving qualitatively *different* amounts of postelection transfers. This situation would arise if the two municipalities were in districts where the degree of asymmetry in municipality size varied. Because prior work did not have the tournament theory from which to draw, it did not appreciate that the relative sizes of the groups into which voters are arranged exerts an independent effect on transfers. Thus, when scholars examined whether municipalities that were more supportive of their LDP incumbent received more transfers, they did not always restrict their comparison to municipalities *in the same district*. They adopted specifications that effectively asked whether, generally speaking, municipalities with higher levels of support received more transfers, and they found that the answer to this question was "no." In Chapter 6, I showed that once the comparison is limited to municipalities *in the same district*, accomplished with a regression specification that includes district-year fixed effects, municipalities with higher levels of electoral support *do* receive more transfers.

This chapter is devoted to empirically examining whether the degree of asymmetry in municipality size exerts an independent effect on the amount of transfers received by municipalities in the district, in the direction predicted by the theory. If, controlling for other differences across electoral districts, the total amount of money each district receives is found to be a function of the degree of asymmetry in sizes of its constituent municipalities, then this is further evidence that LDP politicians are using tournaments to get elected. This piece of evidence helps us discriminate between what Chapter 4 called contest-based compensation (a tournament), in which money is allocated on the basis of rank, and "output-calibrated compensation," in which money is awarded to groups in proportion to the group's contribution to the politician's total vote share in the district. If LDP politicians are using output-calibrated compensation, then we would still observe the positive relationship between electoral support and transfers *within* districts (documented in Chapter 6), but there would be no reason for the total amount of transfers awarded to districts to vary systematically with the relative size of constituent municipalities. Evidence of this is further evidence of contest-based compensation (a tournament).

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7.1 EMPIRICAL STRATEGY

Chapter 4 fleshed out the theoretical underpinnings of the hypotheses this book tests. For ease of access, here are the two examined in this chapter.

- **Hypothesis 3**: Across electoral districts with LDP winners, resource allocations are a function of the degree of asymmetry in municipality size. All else equal, districts characterized by greater asymmetry in municipality size receive larger resource allocations.
- **Hypothesis 4**: Across electoral districts, support for the LDP is a function of the degree of asymmetry in size of municipalities therein. All else equal, electoral districts characterized by greater asymmetry in municipality size exhibit less support for the LDP.

7.1.1 Variables

In Chapter 5, I classified electoral districts used in Japan's Lower House elections as tournament-possible or tournament-impossible. Tournament-possible districts are divisible into more than one municipality, at which electoral support is discernible and to which resources are deliverable. It is in these districts in which LDP politicians can conduct tournaments. Because tournament-impossible districts consist of a single municipality, have sizeable shares of their voters residing in a municipality whose area is not fully contained in the district, or municipalities to which central government resources are not easily deliverable, they do not meet these conditions. Because LDP politicians cannot conduct tournaments in tournament-*impossible* districts, tournament-*possible* districts are the focus of my analyses in these empirical chapters.

As I explained in Chapter 6, my data consist of yearly observations (where t indicates the year) of electoral districts (d), municipalities (m), and candidates (c). Because tournament-possible districts are comprised of multiple self-contained municipalities, the creation of district-level variables capturing the level of support offered the LDP, the amount of money (NTD) received after elections, and other demographic and fiscal characteristics with the data I collected is relatively straightforward and simply requires the aggregation of these municipality-level variables. Below, I provide more detail.

First, I created Winning LDP $\text{VS}_{d,t}$, a variable capturing the level of support electoral district d offered its LDP winners.[4] Let $v_{c,m,t}$ represent the number of votes obtained by candidate c in municipality m in district d at time t. This is available for $t \in E = \{e_1, e_2, \ldots\} =$

[4] Where possible, I use the variable operationalizations in Catalinac, Bueno de Mesquita and Smith (2020).

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{1980, 1983, . . .}, the years in which Lower House elections were held. Let $p_{m,t}$ represent the voting population in municipality m in district d at time t (the number of eligible voters). Several indicator functions can also be defined: let $w_{c,t}$ indicate whether candidate c won a seat in district d at time t, and $LDP_{c,t}$ indicate whether candidate c was a member of the LDP at time t. Because municipality m belongs to a single district d, I calculated, for each district d:

$$\text{Winning LDP VS}_{d,t} = \frac{\sum_{m \in d} \sum_{c \in m} \{LDP_{c,t} w_{c,t} v_{c,m,t}\}}{\sum_{m \in d} p_{m,t}}. \qquad (7.1)$$

Colloquially, Winning LDP $\text{VS}_{d,t}$ captures the proportion of eligible voters in district d at time t who voted for a winning LDP candidate. Put differently, it is the share of available votes in district d at time t that was captured by winning LDP candidates. Note that this operationalization differs from alternative operationalizations in that the numerator is votes cast for *winning* LDP candidates (not all LDP candidates), while the denominator is number of *electors* (not votes cast).

Second, to capture my main independent variable, the degree of asymmetry in municipality size in district d at time t, I constructed a standardized Herfindahl Index, what I call $\text{HI}_{d,t}$. Let $\text{n}_{d,t}$ represent the number of municipalities in each district d at time t. Let $p_{d,t}$ represent the voting population of district d at time t. For each municipality m in district d at time t, I take its voting population ($p_{m,t}$) and divide this by the voting population of its district ($p_{d,t}$) to calculate the fraction of the district's population that lives in municipality m. For all municipalities in the same district, I square each of these fractions and take the sum of these squares. This first step creates an index that runs from 0 to 1, with 1 capturing a district in which every voter lives in a single municipality (the district's voters are concentrated in a single municipality) and lower numbers capturing districts in which voters are divided amongst the municipalities in the district.

Next, I standardize the index so that regardless of the number of municipalities in each district, $\text{HI}_{d,t} = 1$ captures a district in which the voting population is concentrated in a single municipality and $\text{HI}_{d,t} = 0$ captures a district in which the voting population is evenly divided amongst the municipalities in the district.[5] To do this, we take the sum of these squares minus 1 divided by the number of municipalities in

[5] Without standardizing the index by number of municipalities in a district, a district consisting of four municipalities in which the voting population was perfectly divided among each would have a higher score on the index (0.25) than a district consisting of five municipalities in which the voting population was perfectly divided among each (0.20). After standardizing the index by number of municipalities in the district, both of these hypothetical districts would have 0 on the index.

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the district, and divide this result by 1 − 1 divided by the number of municipalities in the district. Concretely:

$$\mathrm{HI}_{d,t} = \frac{\sum_{m \in d} \left(\frac{p_{m,t}}{p_{d,t}}\right)^2 - 1/n_{d,t}}{1 - 1/n_{d,t}}. \tag{7.2}$$

Colloquially, $\mathrm{HI}_{d,t}$, uses variation in the populations of municipalities in district d to capture the degree to which a district's population is concentrated in a single municipality ($\mathrm{HI}_{d,t} = 1$) or spread out evenly across multiple municipalities ($\mathrm{HI}_{d,t} = 0$).

Third, I created Postelection $\mathrm{Transfers}_{d,t}$, a variable capturing the logarithm of the per capita allocation of NTD received by districts in the fiscal years following Lower House elections. To do this, I took all municipalities in district d and summed the amounts of NTD received by them in the fiscal years following each Lower House election. This is the total amount of NTD received by district d in the year following an election held at t. I calculated the district's population analogously, by summing up the populations of the municipalities located within district d. Dividing the total amount of transfers received by district d in the year after the election by the population of district d generates a per capita measure, which I then take the logarithm of.

Fourth, I constructed district-level versions of the six control variables included in Chapter 6's analyses of transfers to municipalities. Population in district d at time t was calculated by summing the populations of municipalities in district d. Population density was calculated by dividing the district's population by its geographic area (in kilometers squared), captured by summing up the geographic area of all municipalities located within it. The fiscal strength of district d at time t was calculated by multiplying each municipality's value on the fiscal strength index by the fraction of the district's population constituted by that municipality, and then summing those for all municipalities in district d. The proportion of residents employed in primary industries; the proportion of residents aged fifteen and under and sixty-five and over; and per capita income at the district-level were created the same way (each municipalities' values were multiplied by the fraction of the district's population constituted by that municipality, and the resulting values were summed for municipalities in the same district).

7.1.2 Potential Confounders

My goal is to examine whether resource allocations across electoral districts vary systematically with the degree of asymmetry in municipality size. Prior work on resource allocations to larger entities – namely,

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electoral districts and prefectures – in Japan have identified other variables thought to influence the amount of money received by these larger entities. Any analyses attempting to establish the importance of a new variable such as district-level asymmetry should control for the potentially confounding effects of these variables. They are malapportionment; the share of a district's seats held by LDP politicians; the fiscal strength of the entity (the degree to which it can finance the cost of services itself or must rely on the central government); the number of municipalities in the district; and whether the district counts a senior LDP politician among its Lower House representatives, respectively. Here, I recount the claims scholars have made about why each of these variables matter and summarize what their empirical analyses showed. I conclude my discussion of each variable by explaining how I operationalize it with my data.

First, Horiuchi and Saito (2003) make the case that malapportionment – inequality in representation – affects the amount of central government transfers received by voters in different electoral districts. Malapportionment captures a situation in which some districts have more Diet Members per voter than others. When a system is malapportioned, the weight of an individual vote is not the same across districts: In some districts, more votes are required to elect a Diet Member than in others. How does malapportionment come about? While electoral districts may have been allotted the "correct" number of Diet members relative to population when their boundaries were first drawn up and the number of seats (their district magnitude) was decided upon, the combination of out- and in-migration over time usually necessitates corrections, either to the number of Diet Members elected in each district or to district boundaries themselves. Seats will need to be added to districts experiencing in-migration (urban districts) and subtracted from districts experiencing out-migration. If such corrections are not carried out, then malapportionment is the result.

In Japan, despite being required to use the results of censuses conducted at five-year intervals to prevent malapportionment from becoming too severe, successive LDP governments were very slow to correct vote-seat disparities (Horiuchi and Saito 2003; Meyer and Naka 1999). Despite malapportionment increasing in every Lower House election held under the party's watch, it managed to conduct "minor reapportionments" only four times between 1955 and 1994, when Japan's electoral reform necessitated a complete redrawing of district boundaries (which dramatically reduced malapportionment). It was not until 1986 and 1993, respectively, that the LDP government actually managed to take seats *away* from certain (rural) districts. Horiuchi and Saito (2003) posited that municipalities in overrepresented districts (those

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with more Diet representatives per voter) would receive more central government transfers. Regression analyses of the determinants of the amount of per capita transfers (the sum of both the formula-based [LAT] and discretionary [NTD] transfer described in Chapter 6) received by Japanese municipalities showed that district-level malapportionment (namely, the number of seats available in the district, divided by the district's population, in logs), had a positive, statistically significant effect on transfers, after controlling for other differences across municipalities. I constructed People Per $\text{Seat}_{d,t}$, which is the district's population divided by the number of seats available.[6]

Second, a variable that was controlled for in the cross-sectional analyses in Horiuchi and Saito (2003), conducted on separate years before and after Japan's electoral reform, was share of a district's seats occupied by LDP-affiliated politicians. Put simply, this is the share of a district's representatives who are LDP-affiliated. Their results showed that this exercised a statistically significant, *negative* impact on the amount of transfers received by Japanese municipalities in the years 1991 through 1993. In the years 1994 through 1998, the coefficient on this variable remained negative but ceased to be statistically significant. Viewed together, their results suggest that Japanese municipalities receive more transfers when their district has more Diet representatives per voter, but *fewer* transfers when more of those representatives are LDP-affiliated.

Other studies found similarly negative or null effects of indicators of electoral support for the LDP at the district-level. McMichael (2018) studied the determinants of the amount of per capita transfers (the sum of LAT and NTD) received by Japanese prefectures in the period 1958–1993. He tallies up the number of Lower House Members elected in each prefecture (electoral districts are perfectly contained within prefectures) and calculates the share of those Diet members who are LDP-affiliated. Various regression specifications, with logged and nonlogged dependent variables, with the full sample and a truncated sample, revealed that the share of a prefecture's Diet seats occupied by LDP politicians exercised either a statistically significant, *negative* impact on the amount of transfers received, or no effect. Saito (2010, Chapter 5) also studied the determinants of transfers to Japan's electoral districts over the period 1977–1990. Controlling for many other differences across electoral districts, he found that both the share of a district's seats held by LDP politicians and the size of the LDP's vote share at the district level had no effect on the amount of per capita transfers (the sum of LAT and

[6] Following Catalinac, Bueno de Mesquita and Smith (2020), I scaled it so that the value represents the number of people, in 100,000s, per available seat in the district.

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NTD) received. In his regression models, the coefficients on these district-level indicators of the LDP's strength were always negative. In addition to Winning LDP $VS_{d,t}$, described above, I calculated LDP $Seats_{d,t}$, which captures the proportion of seats in district *d* that were won by LDP candidates.[7]

Third, Scheiner (2005, 2006) argued that fiscal centralization in Japan created a situation in which regions of the country that were prosperous and had less of a need for government transfers were less tethered to the ruling party than regions that were less prosperous and had a greater need for government transfers. By and large, the less-prosperous regions, characterized by lower incomes and weaker tax bases, were rural areas. There, he observed that local politicians tended to affiliate with the ruling LDP and use their ability to mobilize votes for LDP-affiliated Diet members to make sure those Diet Member favored their regions when deciding on allocations of central government transfers. Every year, the Japanese government makes data on the "fiscal strength" of each prefecture and municipality available. Put simply, this is an index created by dividing the prefecture or municipality's revenue by its needs.[8] Higher scores indicate greater autonomy from the central government, while lower scores indicate less autonomy. Regression analyses of the proportion of prefectural assembly members affiliated with an opposition party in Japan's forty-seven prefectures between 1971 and 1991 as a function of the prefecture's score on this index revealed that prefectures with greater autonomy from the central government (higher scores on the index) had larger shares of opposition-affiliated politicians than prefectures with less autonomy (lower scores on the index). My controls thus include variables capturing the fiscal strength of the district, as well as the proportion of district's population employed in agriculture.

Fourth, Saito (2010, Chapter 5) made the case that, in addition to malapportionment, the number of municipalities in electoral district *d* exercises an independent effect on the amount of transfers received by electoral district *d*. He argued that electoral districts consisting of multiple municipalities received larger transfer allocations than districts consisting of fewer municipalities. Why? His rationale was that electoral districts divided into more municipalities would have more local politicians. Because municipalities depend on the central government for transfers, most local politicians tend to affiliate with the LDP and mobilize votes

[7] In McMichael (2018)'s analysis of transfers to prefectures, he finds that prefectures in which the party lost a seat in the previous election tended to received more transfers. Below, I include specifications that control for this possibility using LDP $Seats_{d,t}$.

[8] For more information on this index, see McMichael (2017).

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for LDP-affiliated Diet Members in Lower House elections. Thus, electoral districts with more local politicians would have larger cadres of vote mobilizers for the LDP. Another feature of electoral districts consisting of multiple municipalities is that each municipality tends to be smaller. In smaller municipalities, he reasoned that those cadres of vote mobilizers would be better positioned to pressure residents to vote LDP. As a result, he posited that electoral districts with more municipalities would produce higher vote shares for the LDP. Regression analyses of the determinants of per capita transfers (the sum of both the formula-based and discretionary transfer) to Japan's electoral districts in the period 1977 through 1990 showed that the number of municipalities in district *d* was found to exercise a positive, statistically significant effect on the transfers it received, controlling for other differences across districts. I constructed Log(Number of Municipalities$_{d,t}$), which is the logarithm of the number of municipalities in each district.

Fifth (and finally), scholars have suggested that the seniority of the LDP politicians in the district might influence the amount of transfers received (McMichael 2018; Reed 2001; Meyer and Naka 1998, 1999). Their reasoning is that politicians with more terms in office would have built stronger connections to the bureaucracy, and given that bureaucrats decide on transfer allocations, stronger connections would translate into more projects approved for their municipalities. Reed (2001, 120) knits together the occurrence of large-scale corruption scandals implicating senior LDP politicians, the fact that a good number of those scandals involved the construction industry, the "massive" amounts spent on construction by the government each year, and the regular surfacing of anecdotes in the media alleging political influence over grant allocation as evidence corroborating this possibility. He writes: "In no country have so many prime ministers been investigated, tried, and even convicted on corruption charges before becoming prime minister . . . and in no country have so many election campaigns focused on corruption scandals" (Reed 2001, 113–114).

To examine the impact of seniority on grant allocation, he compared annual amounts spent on construction in two electoral districts represented by *powerful* LDP politicians to annual amounts spent on construction in seven similar districts represented by *ordinary* LDP politicians. His research design had the advantage of selecting districts that spanned the entire prefecture. The two powerful politicians he selected were successive leaders of the faction headed by Tanaka Kakuei, who remains the most corrupt Prime Minister in Japan's history. The former (Takeshita Noboru) served as Prime Minister and the latter (Kanemaru Shin), who preferred to control things behind the scenes, had to step

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down after becoming embroiled in a massive scandal implicating the construction industry. Reed (2001) writes: "If Takeshita and Kanemaru proved unable to influence the allocation of public funds for their home prefectures, we can rest assured that no politician (with the possible exception of Tanaka himself) could do so."

Ultimately, his analysis revealed no differences in spending across the two sets of electoral districts. On this basis, he concluded that powerful LDP politicians do not influence the allocation of subsidies to local governments and "many of the anecdotes alleging political influence are simply false" (Reed 2001, 125). Echoing this, McMichael (2018) found that a variable capturing the number of terms served by all a prefecture's LDP politicians had no statistically significant relationship with transfers to that prefecture. I constructed Senior LDP Politician$_{d,t}$, which is a dummy variable coded "1" if one of the LDP winners in district d at time t had served at least five terms.[9]

7.2 ASYMMETRIC ELECTORAL DISTRICTS RECEIVE MORE MONEY

Having elucidated the electoral district-specific characteristics others suggest matter, let me return to the theory. The theory expects that the total amount of money LDP politicians will deliver to the groups of voters in their electoral district will be a function of the ease with which tournaments can be administered. Ease, in turn, is a function of the relative sizes of the groups in which voters reside. When groups are *similarly sized*, there will be less ambiguity over how the performance of groups will be compared to each other, so LDP politicians will be able to use relatively *small* prizes to entice groups to compete. When groups are *asymmetrically sized*, in contrast, there will be greater ambiguity over how groups will be compared to each other, so LDP politicians will have to proffer *larger* prizes in order to convince voters to turn out and vote for them. This is Hypothesis 3.

To examine this conjecture, I constructed HI$_{d,t}$, an index capturing, for every electoral district d at time t, the degree to which its population is concentrated in a single municipality (HI$_{d,t}$ = 1) or spread out evenly across multiple municipalities (HI$_{d,t}$ = 0). This is the main independent variable of interest in the analyses that follow. First, let us verify that we have variation in HI$_{d,t}$ across our periods of study. My sample in the analyses in this chapter is electoral districts in which LDP politicians are

[9] As McMichael (2018) and Krauss and Pekkanen (2010) argue, LDP politicians advance to leadership positions in the party, the Diet, or the Cabinet in their fifth term.

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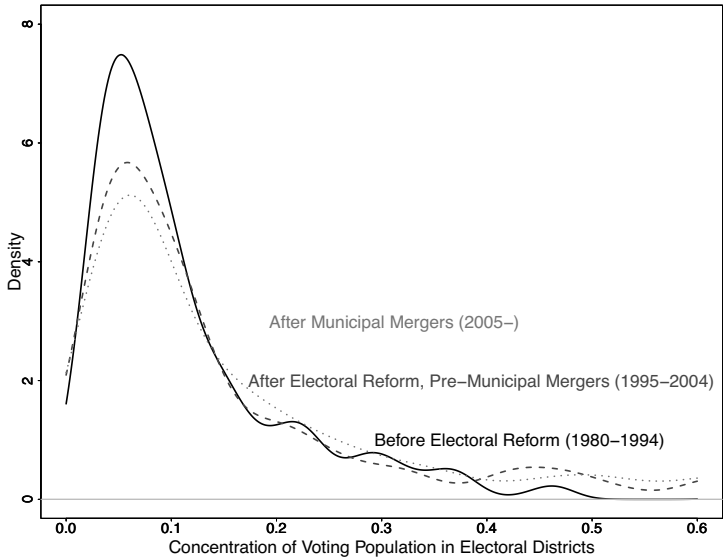

Figure 7.1 A plot of the concentration of voting population per electoral district for tournament-possible districts with LDP winners, 1980–2014.
Higher (lower) values indicate districts where the voting population is more (less) concentrated in a single municipality. The shape of the distribution is similar in all three periods.

conducting tournaments: thus, tournament-possible districts with LDP winners in all Lower House elections, 1980–2014.[10]

Figure 7.1 plots the distribution of voting concentration ($HI_{d,t}$) across this sample of electoral districts in three distinct periods. The black solid line plots the distribution in the 130 (and 129) multiseat districts used in Lower House elections from 1980 until 1993. The dark gray dotted line plots the distribution in the 300 single-seat districts (SSDs) used in the 1996, 2000, and 2003 Lower House elections. For the most part, these elections occurred prior to a large-scale reduction in number of municipalities, which took place over the 2001–2005 period. Finally, the light gray dotted line plots the distribution in the 300 SSDs used in the 2005, 2009, 2012, and 2014 Lower House elections. These elections occurred after many municipalities merged, which led to a substantial reduction in number of municipalities per district. In all three periods, the shape of the distribution is similar: There are fewer districts in which voters are concentrated in a single municipality (higher values of $HI_{d,t}$), and more districts in which voters are spread out across municipalities of relatively

[10] The empirical tests in this chapter were inspired by those in Catalinac, Bueno de Mesquita and Smith (2020).

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similar sizes (lower values of $HI_{d,t}$). Importantly for my purpose, districts with both high and low voting concentration exist in all three periods.

7.2.1 Comparing Electoral Districts in the Same Election

To examine the effects of $HI_{d,t}$, I conduct regressions in which observations are electoral districts, not municipalities. Descriptive statistics pertaining to the district-level variables used in these regressions appear in Table A.8. Table 7.1 presents fixed effect regression models for the logarithm of per capita transfers received by electoral districts in the years following the Lower House elections held between 1980 and 2014 as a function of the concentration of the district's voting population ($HI_{d,t}$), year fixed effects, and other district-level controls. Because Chapter 6 established that tournaments are conducted by winning LDP politicians, observations are limited to districts that are tournament-possible and produced at least a single LDP winner.

In both models, year fixed effects mean that I am leveraging variation in the amounts of transfers received by districts with LDP winners in the same election. Both models also include district-level versions of the six municipality-level controls used in Chapter 6's analysis of transfers to municipalities within districts. One of these is fiscal strength, which prior work suggests will negatively influence transfer amounts. Other controls are operationalizations of the variables found in prior work to influence transfers to districts: malapportionment, number of municipalities in the district, share of seats occupied by LDP politicians, and whether a senior LDP politician represents the district (a dummy variable coded 1 and 0). I report robust standard errors clustered on electoral district.[11]

In Model 1, the coefficient on $HI_{d,t}$ is positive and statistically significant, indicating that in the period 1980–2014, districts characterized by greater asymmetry in municipality size received larger per capita transfer allocations after the election, controlling for other differences across districts, as well as features of a given election that might influence amounts of transfers received by all districts. Substantively, Model 1 tells us that a one standard deviation increase in $HI_{d,t}$ can be expected to net a district an extra 11.5% increase in per capita transfers after the election. Given that the mean per capita transfer to districts for these fiscal years was 30,576 yen, this amounts to a per capita increase of 3,517 yen. The mean district in this sample contained more than 500,000 voters; thus, a

[11] In this and all subsequent regressions in this chapter, I use district-level identifiers that change when a district's boundaries change. I rely on the identifiers in Reed and Smith (2015).

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Table 7.1 *Across districts with at least one LDP winner, greater asymmetry in municipality size ($HI_{d,t}$) is associated with larger per capita transfer allocations after Lower House elections, 1980–2014.*

	Dependent variable: Postelection Transfers (log)	
	All electoral districts	Excludes prefectural capitals
	(Model 1)	(Model 2)
$HI_{d,t}$	0.712***	0.620***
	(0.128)	(0.147)
Fiscal Strength$_{d,t}$	−0.263*	−0.276*
	(0.107)	(0.118)
Dependent Population$_{d,t}$	4.409***	4.041***
	(0.537)	(0.553)
Farming Population$_{d,t}$	3.563***	3.351***
	(0.878)	(0.925)
Log(Population$_{d,t}$)	0.017	−0.048
	(0.107)	(0.120)
Log(Per Capita Income$_{d,t}$)	0.450***	0.405*
	(0.124)	(0.159)
Population Density$_{d,t}$	0.030*	0.028*
	(0.012)	(0.013)
Log(Number of Municipalities$_{d,t}$)	0.176***	0.153***
	(0.044)	(0.046)
People Per Seat$_{d,t}$	−0.078	−0.041
	(0.040)	(0.042)
Winning LDP VS$_{d,t}$	−0.311	−0.170
	(0.179)	(0.193)
LDP Senior Politician$_{d,t}$	−0.042	−0.037
	(0.028)	(0.029)
Observations	1,414	1,151
Year FE	Y	Y
R^2	0.407	0.391

Model 1 uses the full sample of districts, while Model 2 excludes districts in which prefecture capitals are located.

Robust standard errors clustered on electoral district in parentheses.
$^{*}p < 0.05$, $^{***}p < 0.001$.
Regression models estimated with R's plm() package.

per capita increase of 3,517 yen translates into an increase of more than 1.841 billion yen, or almost $12 million in USD in today's exchange rate.

As I explained in Chapter 5, the upper tier of local government in Japan is the prefecture, of which Japan has forty-seven. Each prefecture

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has a capital city, in which the Governor and prefectural assembly sit. For the purpose of Lower House elections, each prefecture is divided into a varied number of electoral districts, one of which houses the prefecture's capital. Electoral districts in which prefectural capitals are located typically consist not only of the large (capital) city but also of the much-smaller towns and villages in the adjacent area. Because prefectural capitals are disproportionately large, these districts may have disproportionately high values of $HI_{d,t}$. This could be a concern if prefectural capitals receive more central government transfers for other reasons, such as enhanced lobbying capacity or closer connections to their LDP-affiliated Diet Members.

Across the twelve elections, I found that the average electoral district in which a prefectural capital is located contained twenty-seven municipalities and had an $HI_{d,t}$ of 0.22. The average electoral district *outside* the prefectural capital contained seventeen municipalities and had a lower $HI_{d,t}$ of 0.13. These differences in means are statistically significant. Thus, electoral districts hosting prefectural capitals *do* tend to be more asymmetric than districts not hosting prefectural capitals. In Model 2 of Table 7.1, I run the same model as in Model 1 but exclude districts in which prefectural capitals are located. The coefficient on $HI_{d,t}$ remains positive and statistically significant. This means that the results are not driven by the inclusion of districts that are characterized by greater asymmetry in municipality size and might receive additional transfers for other reasons. Substantively, according to Model 2, in districts outside the prefecture capital, a one-standard-deviation increase in $HI_{d,t}$ can be expected to net a district an extra 8.9% increase in per capita transfers. The mean per capita transfer to districts in this sample was 31,097 yen, so this amounts to a per capita increase of 2,776 yen.

In both models in Table 7.1, I controlled for Winning LDP $VS_{d,t}$, the share of eligible voters in the district who voted for winning LDP candidates. In both models, its coefficient is negative and not statistically significant. This means that, comparing districts with LDP winners in the same election, the strength of electoral support districts give their LDP winner(s) has no effect on transfers, at least when we are controlling for district-level asymmetry. In the Appendix, we present two additional tables. Table A.9 presents the same two models with LDP $Seats_{d,t}$ (the share of seats in a district won by LDP candidates) and LDP $VS_{d,t}$ (which includes votes for losing LDP candidates in the numerator), respectively.[12] Regardless of how we measure support for the LDP, $HI_{d,t}$ is found to

[12] Note that because we limit observations to electoral districts with LDP winners, LDP $Seats_{d,t}$ ranges from 0.2 to 1 in elections prior to 1994 and is limited to 1 after reform.

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exercise a positive, statistically significant impact on postelection transfers. Table A.10 presents the same specifications in Table 7.1, but without any controls for strength of LDP support at the district-level. We do this because the theory *expects* the relationship between $HI_{d,t}$ and support for the LDP to be endogenous: Increases in $HI_{d,t}$ are expected to increase transfers because they lower the propensity to turn out and vote for the LDP. Thus, including both $HI_{d,t}$ and Winning LDP $VS_{d,t}$ could introduce posttreatment bias. In this model, the coefficients on $HI_{d,t}$ remain positive and statistically significant. Thus, it is unlikely that the coefficients on $HI_{d,t}$ in Table 7.1 are merely capturing the effects of support for the LDP.

7.2.2 *Comparing Electoral Districts over Time*

As I explained in Chapter 5, the introduction of a new electoral system consisting of SSDs in one tier and closed-list proportional representation (CLPR) in another tier in 1994 necessitated the redrawing of the boundaries of Japan's multiseat districts. 130 multiseat districts became 300 SSDs. Only eight of the new districts had the same borders as districts under the old electoral system, leaving 292 districts that were entirely new creations (Reed 1995, 1077).[13] This means that for the most part, electoral districts before and after 1994 are not comparable. The sample of districts in Table 7.1, which spans all Lower House elections held between 1980 and 2014, thus includes a set of districts in the five Lower House elections prior to reform (1980–1993) and then a totally different set of districts in the seven Lower House elections after reform (1996–2014). My use of year fixed effects in those models enabled me to leverage variation in $HI_{d,t}$ across districts within the same election.

I conducted separate sets of analyses that focus on districts in the five elections prior to reform and districts in the seven elections after reform, respectively. I do this for two reasons. One is to exploit the fact that I have stable district boundaries within the first period, which enables me to leverage variation in within-district $HI_{d,t}$ over time. If I find that district-level asymmetry exercises a positive, statistically significant impact on transfers when I leverage variation in district-level asymmetry across districts in the same election (above) *and* within the same district over time (below), this will provide even greater confidence in the results. The second reason I conduct separate analyses on both periods is to see if the

[13] Districts used in the CLPR tier are much larger. There are eleven in total, and these are drawn around the borders of Japan's 47 prefectures. These are not the subject of our analyses in this book.

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positive relationship between $HI_{d,t}$ and transfers observed in Table 7.1 is observable under both electoral systems. The sample in Table 7.1 includes more observations from the post-1994 period than from the pre-1994 period, owing to the fact that the reform increased the number of districts, and also to the fact that my sample includes more elections under the new system. Because other scholars have demonstrated that Japan's electoral reform changed the allocation of transfers, both within and between districts (Hirano 2006; Horiuchi and Saito 2003), it would further increase confidence in my results if we observed the same relationship between district-level asymmetry and transfers, under both electoral systems.

Table 7.2 presents fixed effect regression models for the logarithm of per capita transfers received by electoral districts in the years following the five Lower House elections held between 1980 and 1993 as a function of the concentration of the district's voting population ($HI_{d,t}$), year fixed effects, and other controls. In Model 1, the specification is the same as Model 1 of 7.1 (year fixed effects and district-level controls), with observations restricted to the five elections prior to reform. The coefficient on $HI_{d,t}$ is positive and statistically significant, showing that, looking across districts in the same election, districts characterized by greater asymmetry in municipality size received more transfers. Substantively, Model 1 tells us that a one-standard-deviation increase in $HI_{d,t}$ can be expected to net a district an extra 13.7% in per capita transfers after the election. Given that the mean per capita transfer to districts in these fiscal years was 25,426 yen, this amounts to a per capita increase of 3,487 yen.

In Model 2, I add district fixed effects to the specification in Model 1. District fixed effects control for time-invariant features of districts that might be influencing the amount of transfers they receive after elections. As such, they help guard against the possibility that underlying characteristics of an electoral district, such as the voters who live therein, are driving any observed effect of $HI_{d,t}$ on postelection resource allocations. As I explained in Chapter 6, however, if there is little variation in the independent variable of interest within the same unit over time, then any effects of this variable will be subsumed by the unit fixed effect. In other words, the model will not be able to estimate its effect separately from the district fixed effect. We can expect that within-district change in $HI_{d,t}$ will be relatively slow-moving, as districts have to undergo quite-sizeable compositional shifts in population to effect meaningful changes in their scores. I illustrate this below in a discussion of municipalities in Nara prefecture, but suffice to say, this biases us against finding any effect of $HI_{d,t}$ in a specification with district fixed effects, unless we can observe the same district over many elections. The coefficient on $HI_{d,t}$ in Model 2 is nevertheless statistically significant. This is evidence that as districts

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Table 7.2 *Across electoral districts with at least one LDP winner, greater heterogeneity in municipality size (HI) is associated with receiving more per capita transfers after Lower House elections, 1980–1993.*

	Dependent variable: Postelection Transfers (log)			
	All electoral districts		Excludes prefectural capitals	
	(Model 1)	(Model 2)	(Model 3)	(Model 4)
$\text{HI}_{d,t}$	1.275***	2.336***	1.215**	3.242***
	(0.276)	(0.635)	(0.425)	(0.949)
Fiscal $\text{Strength}_{d,t}$	−0.233	−0.350*	−0.175	−0.212
	(0.203)	(0.136)	(0.199)	(0.188)
Dependent $\text{Population}_{d,t}$	6.881***	0.145	3.963*	−1.055
	(1.611)	(1.044)	(1.731)	(1.298)
Farming $\text{Population}_{d,t}$	3.575**	−1.016	2.342*	−2.655
	(1.313)	(1.177)	(1.004)	(1.439)
Log($\text{Population}_{d,t}$)	0.100	−0.645***	−0.057	−0.988***
	(0.157)	(0.152)	(0.159)	(0.257)
Log(Per Capita $\text{Income}_{d,t}$)	0.338	0.459***	−0.117	0.117
	(0.218)	(0.114)	(0.235)	(0.148)
Population $\text{Density}_{d,t}$	0.045	−0.053*	0.041	−0.038
	(0.030)	(0.021)	(0.028)	(0.046)
Log(Number of Municipalities$_{d,t}$)	0.211	0.973**	0.173	0.489
	(0.111)	(0.304)	(0.128)	(0.407)
People Per $\text{Seat}_{d,t}$	−0.106	−0.020	−0.007	−0.010
	(0.065)	(0.028)	(0.056)	(0.032)
Winning LDP $\text{VS}_{d,t}$	−0.379	0.016	−0.062	0.054
	(0.222)	(0.049)	(0.221)	(0.054)
Senior LDP $\text{Politician}_{d,t}$	−0.112**	−0.004	−0.120**	0.0004
	(0.038)	(0.011)	(0.045)	(0.013)
Observations	578	578	382	382
Year FE	Y	Y	Y	Y
District FE	N	Y	N	Y
R^2	0.470	0.278	0.438	0.267

Models 1 and 2 use all districts, while Models 3 and 4 exclude districts in which prefecture capitals are located. Models 2 and 4 add district fixed effects.

Robust standard errors clustered on electoral district in parentheses.
$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.
Regression models estimated with R's plm() package.

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become more asymmetric, they receive more transfers, controlling for changes in other district-level characteristics.[14]

In Models 3 and 4 of Table 7.2, I continue the focus on elections under the old electoral system (1980–1993) but restrict observations to districts outside prefecture capitals. Model 3 is the same as Model 2 in Table 7.1 (year fixed effects and district-level controls), but with the sample limited to districts outside prefectural capitals in elections prior to reform. The coefficient on $HI_{d,t}$ is positive and statistically significant. Model 4 adds district fixed effects to the specification in Model 3. The coefficient on $HI_{d,t}$ remains positive and statistically significant. In sum, these specifications provide evidence that in elections prior to electoral reform, among LDP-won, tournament-possible districts, district-level asymmetry exercises an independent effect on transfers.

In the models in Table 7.2, I controlled for Winning LDP $VS_{d,t}$. Its coefficient is not significant in any model. Its sign is negative with year fixed effects (Models 1 and 3) and positive with year and district fixed effects (Models 2 and 4). In Models 2 and 4, the coefficient on Winning LDP $VS_{d,t}$ captures the effect of within-district change in strength of support for the LDP. Thus, in contrast to what McMichael (2018) found at the prefecture level, a waning of support for the party within districts does not appear to have triggered an influx of transfers. In Tables A.11 and A.12, I present the same four models, replacing Winning LDP $VS_{d,t}$ with LDP $Seats_{d,t}$ and LDP $VS_{d,t}$, respectively. Results are strikingly similar to those in Table 7.2: $HI_{d,t}$ is always found to exercise a positive, statistically significant impact on transfers after the election in all models. Thus, no matter how electoral support for the LDP is measured, $HI_{d,t}$ is found to exercise a positive, statistically significant impact on transfers after the election in elections prior to reform.

Next, I run similar models on observations from the seven elections following reform. Table 7.3 presents fixed effect regression models for the logarithm of per capita transfers received by electoral districts in the years following the seven Lower House elections post-reform, as a function of the concentration of the district's voting population ($HI_{d,t}$), year

[14] In regression models with the full sample (districts from all twelve elections in Table 7.1), the addition of a district fixed effect renders $HI_{d,t}$ positive but not statistically significant. In this sample, half the observations are districts prior to electoral reform (which exhibit meaningful variation), but the other half are districts in the postreform period. I discuss this further below, but many districts created in 1994 underwent dramatic changes to their boundaries and municipal composition right in the middle of the post-electoral reform period. The lack of statistical significance in a specification with district fixed effects with the full sample could reflect the fact that many districts in the postreform era were not around long enough to estimate their effect separately from the district fixed effect.

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Table 7.3 *Across districts with at least one LDP winner, greater heterogeneity in municipality size (HI) is associated with receiving more per capita transfers after Lower House elections, 1996–2014.*

	Dependent variable: Postelection Transfers (log)	
	All electoral districts	Excludes prefectural capitals
	(Model 1)	(Model 2)
$HI_{d,t}$	0.510***	0.525**
	(0.147)	(0.162)
Fiscal $Strength_{d,t}$	−0.473***	−0.570***
	(0.131)	(0.155)
Dependent $Population_{d,t}$	4.174***	3.775***
	(0.620)	(0.646)
Farming $Population_{d,t}$	3.692**	3.825**
	(1.318)	(1.325)
Log($Population_{d,t}$)	0.185	0.045
	(0.250)	(0.272)
Log(Per Capita $Income_{d,t}$)	0.634***	0.708***
	(0.131)	(0.139)
Population $Density_{d,t}$	0.029*	0.025*
	(0.012)	(0.012)
Log(Number of $Municipalities_{d,t}$)	0.099	0.091
	(0.051)	(0.055)
People Per $Seat_{d,t}$	−0.126	−0.073
	(0.080)	(0.087)
Winning LDP $VS_{d,t}$	−0.380	−0.133
	(0.355)	(0.370)
Senior LDP $Politician_{d,t}$	0.006	0.014
	(0.035)	(0.034)
Observations	836	769
Year FE	Y	Y
R^2	0.414	0.437

Model 1 uses all districts, while Model 2 excludes districts in which prefecture capitals are located.

Robust standard errors clustered on electoral district in parentheses.
$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.
Regression models estimated with R's plm() package.

fixed effects, and other time-varying controls. In Model 1, the coefficient on $HI_{d,t}$ is positive and statistically significant, showing that, controlling for idiosyncratic features of any given election that could influence the amount of transfers received by all districts in that election, districts

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characterized by greater asymmetry in municipality size received more transfers. Substantively, Model 1 tells us that a one-standard-deviation increase in $HI_{d,t}$ can be expected to net a district an extra 9.5% in per capita transfers after the election. Given that the mean per capita transfer to districts in these fiscal years was 34,733 yen, this amounts to a per capita increase of 3,284 yen. In Model 2, we run the same model as in Model 1, but on a sample of districts that excludes prefectural capitals. The coefficient on $HI_{d,t}$ remains positive and statistically significant.

In my analysis of the 1980–1993 period (Table 7.2), I used district fixed effects to control for the possibility that time-invariant features of electoral districts could be driving the observed relationship between asymmetry and transfers. The postelectoral reform period, however, does not offer us a similarly large set of electoral districts whose boundaries and internal composition are constant across consecutive elections. As I explained in Chapter 5, 91% of all electoral districts were tournament-possible under the old electoral system and almost all of them had at least one LDP winner. Moreover, we observe this set of districts across five elections. After electoral reform, the share of tournament-possible districts dropped to 73% and then kept dropping, reaching about 50%. In addition, two sets of redistricting occurred (in 2003 and 2014), in which districts were eliminated in some prefectures and created in others (The Japan Times 2002; Pekkanen, Reed and Smith 2016). For districts affected by the 2003 redistricting, then, we have two observations until 2003, in 1996 and 2000, respectively. For districts created in 2003, we have a maximum of four observations (in 2005, 2009, 2012, and 2014, respectively), unless they were also affected by the 2014 redistricting, in which case we have fewer. While this leaves a set of tournament-possible districts unaffected by redistricting, many of these districts saw their internal composition change decisively due to mergers of municipalities located within them. Thus, while a district's borders (and official code) can be the same across two elections, the district may have undergone a major change in the arrangement of voters into municipalities. In short, features of the post-1994 era suggests we should be cautious about using district fixed effects.

The models in Table 7.3 included Winning LDP $VS_{d,t}$ as a control. Its coefficient is negative and statistically insignificant in both models. Thus, comparing districts in the same election with LDP winners, the strength of support voters offered this winner is found to have no relationship with the amount of transfers it receives.[15] In sum, district-level asymmetry is found to exercise a positive, statistically significant impact

[15] There is no need to examine alternative operationalizations of support for the LDP because the fact that the postreform era uses SSDs and I limit observations

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on the amount of transfers received by districts in the years following Lower House elections, both before and after electoral reform.

7.2.3 Japan's 1994 Electoral Reform as a Quasiexperiment

Japan's 1994 electoral reform entailed the drawing up of 300 new districts. Thus, districts before and after 1994 are not comparable. For Japan's 3,300+ municipalities, the redrawing of district boundaries meant being placed in an entirely new district, consisting of a different mix of municipalities. Because the new districts were *smaller* than the old districts, many new districts were composed of a subset of municipalities from the older, larger district. Thus, in the vast majority of cases, municipalities did not find themselves facing an entirely new set of competitors in the tournament but a set they were already familiar with. Nevertheless, the creation of a new district from a subset of municipalities in the old district means that, for most municipalities, my variable of interest – the degree of asymmetry in municipality size – decisively changed.

Let me illustrate this with two municipalities, Nara City (population 359,604 in 1996) and Tsukigase Village (population 2,028 in 1996), both in Nara prefecture. In the five elections held between 1980 and 1993, these municipalities were in the old Nara First District. Comprised of forty-seven municipalities, this district included ten cities (with populations ranging from approximately 350,000 to 36,000), twenty towns (populations from 28,000 to 5,000), and seventeen villages (populations from 6,000 to 700). This large district sent five representatives to the Lower House. In 1980, its $HI_{d,t}$ was 0.067, right at the median for electoral districts in this period. Over time, it became slightly more asymmetrical, as the populations of its cities increased and the populations of its smaller municipalities declined. By 1993, its $HI_{d,t}$ score was 0.074, which amounts to small increase of 0.007.

The 1994 reform broke up the old district, like it did all districts. Nara City and Tsukigase Village were the only municipalities placed in the new Nara First. Nine municipalities went into Nara Second District, twelve municipalities went into Nara Third, and twenty four municipalities went into Nara Fourth. The asymmetry in size between Nara City and Tsukigase Village, with the former's population being 175 times the latter, meant that the new Nara First had the largest $HI_{d,t}$ of any district in 1996: 0.977. Thus, after having experienced little change in $HI_{d,t}$ between 1980 and 1993, Nara City and Tsukigase Village both experienced dramatic increases in $HI_{d,t}$ between the 1993 and 1996 election, going from $HI_{d,t}$ =

to districts with LDP winners mean that LDP $Seats_{d,t}$ has the value of 1 for all observations and LDP $VS_{d,t}$ is identical to Winning LDP $VS_{d,t}$.

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0.074 to 0.977. The nine municipalities that went into Nara Second also experienced increases in $HI_{d,t}$ (going from 0.074 in 1993 to 0.10 in 1996) and so did the twenty-four municipalities going into Nara Fourth (going from 0.074 to 0.137). In contrast, the twelve municipalities that went into Nara Third experienced a decrease (going from 0.074 to 0.036).

This experience, of very little change in $HI_{d,t}$ in the thirteen years between 1980 and 1993 and then an abrupt change in $HI_{d,t}$ between 1993 and 1996, is reflective of the broader population of Japanese municipalities. For every municipality in my data, I took their $HI_{d,t}$ scores in the five elections between 1980 and 1993 and calculated the standard deviation in these scores. The average standard deviation provides an indicator of the degree of change in $HI_{d,t}$ experienced by the average Japanese municipality over these thirteen years. It is 0.006. I then did the same for every municipality in the 1993 and 1996 elections (took their two $HI_{d,t}$ scores, calculated the standard deviation in these, and then took the mean). This showed that the average municipality experienced a change in $HI_{d,t}$ of 0.030 between 1993 and 1996, five times larger than the change experienced over the preceding thirteen years.

The reason municipalities experienced little change in $HI_{d,t}$ between 1980 and 1993 is because district boundaries were stable. While there were "minor reapportionments" in 1986 and 1993, neither involved the redrawing of district boundaries. Each merely entailed the adding of seats to some districts and the subtracting of seats from others (Horiuchi and Saito 2003, 672). This means that virtually all municipalities squared off against the *same* set of competitors at every election.[16] In this setting, changes in district-level asymmetry were driven by changes in the *populations* of the municipalities comprising the district. Most commonly, residents of smaller municipalities relocated to larger ones, either in the same electoral district or in another one. In the former case, this pattern of migration would lead to an increase in district-level asymmetry. In the latter case, it would likely result in changes to the degree of asymmetry in *both* districts, depending on the relative sizes of the municipalities from which and to which they moved, relative to other municipalities in the district. Critically, however, population shifts along these lines do not have the same impact on $HI_{d,t}$ as being plucked from one district, with one set of competitors, and placed in another, with a different set of competitors.

I can take advantage of these abrupt changes in $HI_{d,t}$ to conduct another test of Hypothesis 3. Specifically, I can examine the effect of changes in district-level asymmetry between the 1993 and 1996 elections on change in the per capita transfers received in the years following

[16] I say "virtually all" because there were a small number of *municipalities* whose borders changed during this time.

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these two elections for all municipalities present in both years. If my hypothesis about the effects of district-level asymmetry is correct, municipalities that experienced greater increases in district-level asymmetry between the two elections will have received a greater increase in their transfer allocation (over and above what they received the year after the previous election) than municipalities that experienced smaller increases in district-level asymmetry.

To do this, I took the universe of municipalities in tournament-possible districts present in the 1996 election and calculated ΔPostelection Transfers$_{m,t}$ for each. This is the logarithm of the per capita transfers received in the year after the 1996 election, minus the logarithm of the per capita transfers received in the year after the 1993 election. This is the dependent variable in the following regressions. My main independent variable is $\Delta HI_{d,t}$, which is the municipality's $HI_{d,t}$ score in the 1996 election, minus its $HI_{d,t}$ score in 1993. To control for the possibility that changes in other municipality-level attributes could be driving any observed effect of $\Delta HI_{d,t}$, I constructed analogous Δ scores for my six municipality-level control variables, by taking each municipality's value in 1996 and subtracting from this its value in 1993. I also constructed ΔSum LDP VS$_{m,t}$ analogously, by taking a municipality's value on Sum LDP VS$_{m,t}$ in the 1996 election and subtracting its value in the 1993 election. Developed in Chapter 6, Sum LDP VS$_{m,t}$ is my preferred indicator of electoral support at the municipality level. It captures the share of available votes in municipality *m* that were cast for the district's LDP winner(s).

In Table 7.4, Model 1 presents linear regressions of ΔPostelection Transfers$_{m,t}$ as a function of $\Delta HI_{d,t}$, prior transfers, support for the LDP winner in the 1996 election, changes in other municipality-level attributes, and fixed effects for the municipality's district in 1996. Chapter 6's municipality-level analyses showed that the main determinants of transfer allocations after elections are a municipality's relative level of support for the LDP and the transfers it received the year of the election, respectively. Thus, the model includes both: a municipality's Sum LDP VS$_{m,t}$ measured in 1996 and the logarithm of the municipality's per capita transfer allocation the year of the election (1996). My use of district fixed effects enables me to control for features of the municipality's district in 1996 that could have consequences for the amounts of transfers received by all municipalities in the same district. As I mentioned, these could include district-level versions of the same municipality-level variables, such as fiscal strength, as well as malapportionment, number of municipalities in the district, share of seats occupied by LDP politicians, and characteristics of those LDP politicians.

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Table 7.4 *Increases in district-level asymmetry between 1993 and 1996 are associated with increases in postelection transfer allocations, controlling for prior transfers, change in support for the LDP winner and other municipality-level attributes, change in district-level attributes, and features of their electoral districts in 1996.*

	Dependent variable: ΔPostelection Transfers$_{m,t}$	
	(Model 1)	(Model 2)
Δ HI$_{d,t}$	0.550*	1.106*
	(0.238)	(0.433)
Sum LDP VS$_{m,t}$	−0.129	−0.130
	(0.136)	(0.137)
Δ Sum LDP VS$_{m,t}$	−0.197	−0.192
	(0.144)	(0.151)
Log(Transfers$_{m,t}$)	0.039	0.038
	(0.022)	(0.022)
Δ Fiscal Strength$_{m,t}$	−0.174	−0.182
	(0.193)	(0.193)
Δ Dependent Population$_{m,t}$	2.700	2.816
	(9.326)	(9.353)
Δ Farming Population$_{m,t}$	1.472	1.738
	(5.755)	(5.798)
Δ Log(Population$_{m,t}$)	1.089	1.137
	(3.394)	(3.400)
Δ Log(Per Capita Income$_{m,t}$)	−0.275	−0.306
	(0.250)	(0.251)
Δ Population Density$_{m,t}$	−0.537	−0.542
	(0.645)	(0.646)
Δ Log(Number of Municipalities$_{d,t}$)		0.285
		(0.233)
Δ People Per Seat$_{d,t}$		−0.043
		(0.110)
Δ Senior LDP Politician$_{d,t}$		−0.107
		(0.105)
Δ LDP Seats$_{d,t}$		0.360
		(0.345)
Constant	−0.016	0.115
	(0.157)	(0.347)
Observations	2,108	2,108
Fixed Effects for 1996 District	Y	Y
R^2	0.090	0.092

*$p < 0.05$.

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Importantly, if all municipalities in the same 1996 district had come from the *same* 1993 district, all would have identical values of $\Delta HI_{d,t}$, which means I would not be able to estimate its effect separately from that of the 1996 district fixed effect. My inclusion of both $\Delta HI_{d,t}$ and a district fixed effect is possible because some districts in 1996 consist of municipalities that originated from *different* 1993 districts. While these municipalities all ended up in the same district in 1996, the fact that some came from one 1993 district and others came from another 1993 district means that they have *distinct* $\Delta HI_{d,t}$ scores. The presence of this "mixing" in my set of 1996 districts is what enables me to estimate the effects of different degrees of change in district-level asymmetry between the two elections in a specification that includes a fixed effect for the municipality's district in 1996. In this specification, then, the coefficient on $\Delta HI_{d,t}$ will tell us whether, of the set of municipalities that ended up in the same district in 1996, those whose $\Delta HI_{d,t}$ scores are larger, meaning they experienced a greater increase in district-level asymmetry when they moved into the new district, also experienced a greater increase in their postelection transfer allocation.

The coefficient on $\Delta HI_{d,t}$ in Model 1 of Table 7.4 is positive and statistically significant. This means that municipalities that experienced a greater increase in district-level asymmetry between the 1993 and 1996 election than their same-district counterparts saw a greater increase in their postelection transfer allocation, controlling for changes in other municipality-level attributes, as well as the absolute level of support the municipality offered the LDP in 1996 and the amount of transfers it received in 1996. My inclusion of ΔSum LDP $VS_{m,t}$ as a control variable means that my results are telling us that regardless of whether a municipality's support for the LDP went up or down between the two elections, municipalities shifted into a district characterized by greater asymmetry than their old district experienced a bigger increase in their postelection transfer allocation.

The coefficient on the amount of transfers the municipality received the year of the election is positive and statistically insignificant. This means that receiving more transfers in the 1996 fiscal year is not associated with experiencing a larger change in transfers between 1994 and 1997, respectively. Similarly, neither the absolute *level* of support the municipality offered the LDP in 1996, nor the degree of *change* in its support between the two elections, is found to be significantly related to change in its postelection transfer allocation. This suggests that the main determinant of change in a municipality's postelection transfer allocation is not electoral support or change in other time-varying municipality-level attributes, but change in district-level asymmetry. This is powerful evidence that

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municipalities in districts characterized by greater asymmetry in municipality size receive more transfers.

In Model 1 of Table 7.4, my use of a fixed effect for the municipality's 1996 district means that I am leveraging variation in $\Delta HI_{d,t}$ across municipalities in the same 1996 district. Having come from different 1993 districts, however, these municipalities will also vary in the degree of change they experienced in *other* district-level features. Model 1 controls for changes in *municipality*-level attributes between the two elections, but not in *district*-level attributes. For example, while most municipalities would have experienced a reduction in the degree to which their district was malapportioned between 1993 and 1996, there could be variation across municipalities in the same 1996 district as to the extent to which malapportionment had been reduced, relative to their old district. Horiuchi and Saito (2003)'s findings suggest that municipalities that experienced greater reductions in malapportionment between the two elections experienced a larger drop in their transfer allocation. Similarly, because the 1996 districts are smaller than the 1993 districts, most municipalities would have also experienced a reduction in the number of municipalities in their district (and correspondingly, in the number of local politicians in their district). But there could be variation across municipalities in the same 1996 district as to the extent to which they experienced a reduction in the number of municipalities. Saito (2010) gives us reason to believe that municipalities experiencing greater declines in number of municipalities in their district will experience larger declines in their transfer allocation.

In Model 2 of Table 7.4, I add controls for change in five district-level features between 1993 and 1996 that could also have consequences for change in municipality *m*'s postelection transfer allocation. I keep everything else about the specification identical to that in Model 1. The inclusion of these variables helps me verify that the effects of $\Delta HI_{d,t}$ in Model 1 are not simply capturing the effects of change in *other* district-level attributes between 1993 and 1996 that could also have consequences for change in a municipality's transfer allocation. Specifically, for each municipality present in 1996, I constructed Δ People Per Seat$_{d,t}$, Δ log(Number of Municipalities$_{d,t}$), Δ LDP Seats$_{d,t}$, Δ District Magnitude$_{d,t}$, and Δ LDP Senior Politician$_{d,t}$. This involves taking the municipality's value on each variable in 1996 and subtracting its value in 1993. We find that the coefficient on $\Delta HI_{d,t}$ remains positive and statistically significant and is larger than in Model 1.[17] This reduces

[17] Δ District Magnitude$_{d,t}$ is not estimated separately in this regression, which means that when a 1996 district has municipalities from different 1993 districts, those

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any concern that the effects of $\Delta HI_{d,t}$ are driven by changes in other district-level characteristics.

In sum, the analyses in this section constitute powerful evidence for Hypothesis 3. Controlling for other differences across electoral districts, including in support for LDP winner(s), and differences across elections, districts characterized by greater asymmetry in municipality size are found to have received more money than districts characterized by less asymmetry in municipality size (Tables 7.1–7.3). Moreover, we found that as districts become more asymmetric, they receive more money, controlling for other changes, including in strength of support for their LDP winner(s) (Table 7.2, Model 2). This means that district-level asymmetry exercises an independent effect on the amount of money Japanese municipalities receive. We saw this even more clearly when we studied what happened when municipalities were plucked from one district in 1993 and placed into a different district in 1996: Municipalities shifted into districts characterized by greater asymmetry in municipality size than their old district experienced an increase in transfers, controlling for changes in a host of other variables, including in support for the LDP (Table 7.4). This is powerful evidence that the same municipality receives more money for a given level of support when its district is characterized by greater asymmetry in municipality size.

7.3 ASYMMETRIC ELECTORAL DISTRICTS EXHIBIT LESS SUPPORT

Section 7.2 established that districts characterized by greater asymmetry in municipality size receive more money. As I explained above, the tournament theory would explain this with the following logic: In a tournament, incumbents structure prizes such that they become exponentially larger at higher ranks. This creates neck-and-neck competition for these higher ranks, which increases the amount of influence wielded by a single vote. Whereas a single vote is never likely to influence who wins the election, it could influence where one's group ends up in the ranking. If increases in rank translate into exponentially larger increases in rewards, then this means that individual votes can wield considerable weight over how much money one's group receives.

However, all of this hinges on voters realizing that the incumbent plans to compare the performance of their group with the performance of other groups in the same district. Incumbents have difficulty making this explicit without drawing attention to the fact that tournaments

different 1993 districts had identical district magnitudes (and were captured by the 1996 district fixed effect).

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turn the accountability relationship central to the democratic process on its head. As Chapter 4 explains, it turns out that when groups are asymmetrically sized, an incumbent who ranks groups on the basis of *number* of votes received could end up with a qualitatively different ranking than an incumbent who ranks groups on the basis of the *share* of voters who voted for them. Because different metrics of comparison lead to different rankings, and because incumbents have difficulty clarifying which metric they intend to use, voters in districts characterized by asymmetrically sized groups will find it more difficult to calibrate the degree of influence their vote holds over the size of their group's prize. Being unable to calibrate this influence translates into a lower propensity to turn out and support the incumbent. Anticipating this, incumbents have incentives to increase the size of the prize delivered to groups in districts characterized by greater asymmetry, to encourage them to turn out in spite of the murkiness surrounding how groups will be compared. This is Hypothesis 3.

What are the implications for electoral support? The theory expects that district-level asymmetry will exercise such a pernicious impact on voter willingness to turn out and support the incumbent that even though the incumbent will try to offset this reduced propensity to turn out and support her with larger prizes, it is unlikely she will be able to offer a prize large enough to completely offset this. This is because incumbents also prefer to minimize the amount of government resources (not to mention time devoted to lobbying) that must be diverted to the generation of prizes. Thus, while incumbents in asymmetric districts *could* deliver a prize large enough to increase the level of support they receive to the level enjoyed by their counterparts in less asymmetric districts, they will also be cognizant of the hefty commitment of resources (and time) that this entails. Thus, another implication of the theory is that incumbents in districts characterized by greater asymmetry in municipality size will be delivering more money (Hypothesis 3) but, in spite of this, will still receive lower levels of electoral support (Hypothesis 4).

In this way, the tournament theory provides an explanation for why, across districts, electoral support for the LDP is often found to be *negatively* correlated with the amount of transfers received. It explains this puzzle in the following way: Incumbents find it harder to get support in districts characterized by greater asymmetry in municipality size. Thus, in these districts, they have to offer more (in terms of money) but will get less (in terms of support). Across districts, then, it will look like politicians are engaging in the curious activity of *rewarding* districts for lower levels of electoral support and *penalizing* districts for high levels of electoral support. Or it may look like they are using resources to court relatively

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unsupportive voters. But they are doing no such thing. Instead, a previously *unnoticed* characteristic of the electoral district (asymmetry in municipality size) is exercising an *independent* effect on electoral support. It is lowering support in some districts and increasing it in others. Politicians, for their part, are *anticipating* the effects of this variable. They know that in highly asymmetric districts, they have to deliver more, while in less asymmetric districts, they can deliver less. This leads to a pattern of spending across districts that is the *inverse* of the pattern of spending within districts: Across districts, politicians deliver the most money to districts where they got the least support and the least money to districts where they get the most support.

To examine Hypothesis 4, I turn to analyses with electoral support for the LDP as the dependent variable (not transfers) and district-level asymmetry ($HI_{d,t}$) as the independent variable. Whereas Tables 7.1–7.3 showed that district-level asymmetry increases transfers, my question here is "Does it also depress levels of electoral support for the LDP?"

Table 7.5 presents fixed effect regression models of electoral support for the LDP in Lower House elections from 1980 until 2014 as a function of the concentration of the district's voting population ($HI_{d,t}$), year fixed effects, and other time-varying district-level controls. Whereas the sample in the analyses of transfers above was limited to tournament-possible districts with LDP winners, here we broaden the sample to include all tournament-possible districts. My three dependent variables are Winning LDP $VS_{d,t}$, a variable capturing the amount of support the district offered its LDP winner(s) (Model 1), LDP $VS_{d,t}$, a variable capturing the amount of support the district offered all LDP candidate(s) (Model 2), and LDP $Seats_{d,t}$, the share of seats available in the district that were captured by LDP winners (Model 3). Year fixed effects mean that I am leveraging variation in the degree of asymmetry in municipality size across districts in the same election. In addition to district-level versions of the six demographic controls used throughout this book, I also control for district magnitude, malapportionment, the number of municipalities in the district, whether there is a senior LDP politician in the district, and the number of LDP candidates running in the district. I report robust standard errors clustered on electoral district.

I add the latter control, for the number of LDP candidates, because of the well-known observation that under Japan's old electoral system, when more LDP candidates ran in a district, LDP supporters risked spreading their votes too thinly amongst those candidates, which on occasion resulted in fewer of them getting elected than if voters had concentrated their votes on a select few LDP candidates. This situation arose because Japan's electoral system conferred a single, nontransferable vote

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Table 7.5 *Across districts, greater heterogeneity in municipality size (HI) is associated with lower levels of electoral support for the LDP in Lower House elections, 1980–2014.*

	Dependent variables		
	Winning LDP $VS_{d,t}$	LDP $VS_{d,t}$	LDP $Seats_{d,t}$
	(Model 1)	(Model 2)	(Model 3)
$HI_{d,t}$	−0.073*	−0.043*	−0.204*
	(0.030)	(0.017)	(0.084)
Fiscal $Strength_{d,t}$	−0.051**	−0.043***	−0.187***
	(0.017)	(0.012)	(0.053)
Dependent $Population_{d,t}$	−0.022	0.065	−0.259
	(0.080)	(0.045)	(0.229)
Farming $Population_{d,t}$	0.438*	0.250	0.350
	(0.203)	(0.137)	(0.535)
Log($Population_{d,t}$)	0.037	0.015	0.281*
	(0.056)	(0.039)	(0.139)
Log(Per Capita $Income_{d,t}$)	0.039***	0.033***	0.109**
	(0.011)	(0.008)	(0.036)
Population $Density_{d,t}$	−0.008***	−0.007***	−0.025***
	(0.002)	(0.001)	(0.006)
Total $Seats_{d,t}$	−0.063***	−0.082***	−0.215***
	(0.016)	(0.012)	(0.038)
Log(Number of $Municipalities_{d,t}$)	0.00002	0.002	−0.032
	(0.011)	(0.006)	(0.031)
People Per $Seat_{d,t}$	−0.029	−0.018	−0.091
	(0.019)	(0.012)	(0.051)
Senior LDP $Politician_{d,t}$	0.105***	0.046***	0.263***
	(0.007)	(0.004)	(0.019)
Number LDP $Candidates_{d,t}$	0.080***	0.130***	0.194***
	(0.007)	(0.007)	(0.017)
Observations	1,877	1,877	1,877
Year FE	Y	Y	Y
R^2	0.332	0.618	0.211

Robust standard errors clustered on electoral district in parentheses.
$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.
Regression models estimated with R's plm() package.

on voters, but more than one candidate was elected in each district. This meant that votes could not be transferred to another candidate of the same party should a voter have voted for a candidate who secured enough votes to win. In such a system, parties can suffer from running too many candidates (Cox and Niou 1994). Thus, districts with more LDP candidates may have cast more *votes* for LDP candidates, but if more of

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those votes went to LDP candidates who *lost* the election, then those districts might have arbitrarily smaller Winning LDP $VS_{d,t}$ and LDP $Seats_{d,t}$ scores, respectively. Controlling for the number of LDP candidates in a district ensures that errors of this type are not confounding my ability to estimate the effect of district-level asymmetry on support for the party.

In all three models, the coefficient on $HI_{d,t}$ is negative and statistically significant. This means that districts characterized by greater asymmetry in municipality size exhibit *less* support for the LDP, controlling for other differences across districts and features specific to a given election. Substantively, Model 1 tells us that a one-standard-deviation increase in $HI_{d,t}$ is expected to lower the district's level of electoral support for LDP winners by 1.24%. The sample mean of Winning LDP $VS_{d,t}$ was 0.25, meaning that 25% of eligible voters in the average district cast their ballots for winning LDP candidates. Model 2 tells us that a one-standard-deviation increase in $HI_{d,t}$ is expected to lower the district's level of support for LDP candidates by 0.73%. The sample mean of LDP $VS_{d,t}$ was 0.31, meaning that 31% of eligible voters in the average district cast their ballots for LDP candidates. Model 3 tells us that a one-standard-deviation increase in $HI_{d,t}$ is expected to lower the share of a district's seats won by the LDP by 3.5%. The sample mean of LDP $Seats_{d,t}$ was 0.62, meaning that on average, 62% of the seats available in a district were won by LDP candidates. This is evidence for Hypothesis 4: District-level asymmetry depresses support for the LDP.

Dividing up the sample into preelectoral and postelectoral reform reveals that the negative, statistically significant coefficient on $HI_{d,t}$ in the models in Table 7.5 is a feature of elections under the *new* electoral system, but not under the old. If we confine observations to districts under the old electoral system (1980–1993) and run the same models, we find that districts characterized by greater asymmetry in municipality size do *not* exhibit systematically lower levels of support for the LDP, whether support is measured in share of voters who voted for LDP winners, share of voters who voted for LDP candidates, or share of seats won by the LDP. The coefficient on $HI_{d,t}$ is negative but not statistically significant in any model (see Table A.13). However, running the same models on districts from the seven elections in the postelectoral reform period produces coefficients on $HI_{d,t}$ that are negative and statistically significant (see Table A.14). Thus, district-level asymmetry leads to noticeably lower support rates for the LDP in elections after reform, but not in elections before reform.[18]

[18] Let me add the caveat that, if we conduct regressions with *municipalities* as observations, not electoral districts, then we observe $HI_{d,t}$ exercising a negative, statistically

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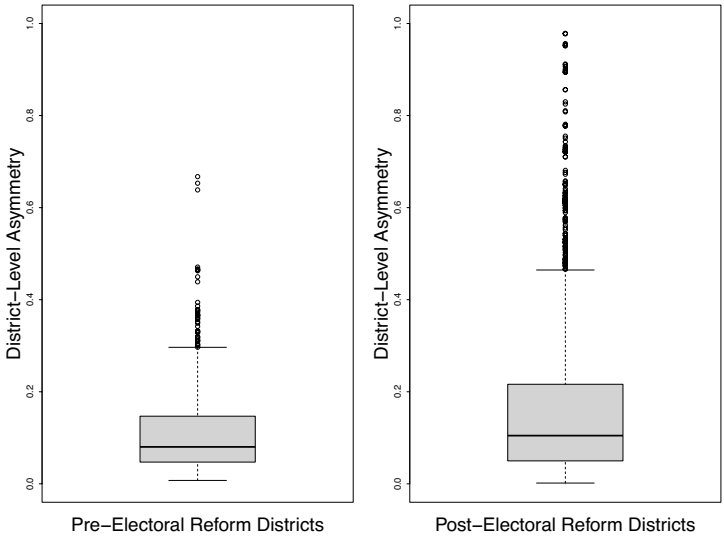

Figure 7.2 This plots the concentration of voting population ($HI_{d,t}$) in districts prior to Japan's electoral reform in 1993 (on the left) and in districts after reform (on the right). Higher (lower) values indicate districts where the voting population is more (less) concentrated in a single municipality.

To understand what could account for this difference, recall that the theory expects that we would not observe districts characterized by greater asymmetry in municipality size exhibiting less support for the LDP if the LDP politicians competing in them had delivered prizes large enough to offset voters' reduced propensity to turn out and support them. The fact that we do not observe a statistically significant, negative correlation between district-level asymmetry and support for the LDP in the prereform period could reflect the fact that LDP politicians were delivering enough money to these districts to increase their support to a level indistinguishable to those of their counterparts in other districts. The fact that we observe a statistically significant, negative correlation between district-level asymmetry and support for the LDP in the postreform period could indicate that LDP politicians in this period were less able to deliver the sums necessary to raise their level of support.

A possible reason for this could be that the redrawing of electoral districts in 1994 had the effect of *increasing* district-level asymmetry. Figure 7.2 plots the distribution of $HI_{d,t}$ across tournament-possible districts in the 1980–1993 elections ($n = 586$ districts) and in the 1996–2014 elections ($n = 1,291$ districts), respectively. A Kolmogorov–Smirnov test confirms that the distributions are significantly different

significant effect on all measures of support for the LDP in both periods, controlling for many other district- and municipality-level characteristics.

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($p>0.001$). Before electoral reform, mean $HI_{d,t}$ across districts was 0.11, only one-quarter of districts had values of 0.15 or higher, and the most asymmetric district was at 0.67. In contrast, after electoral reform, mean $HI_{d,t}$ was 0.18, one-quarter of districts had values of 0.22 or higher, and the most asymmetric district was at 0.98. In 1993, the most asymmetric district was Ehime First District ($HI_{d,t} = 0.67$). It was made up of ten municipalities, which included one large city (Matsuyama City, population = 452,057), one smaller city (Hojo City, population = 29,315), five towns (with populations ranging from approximately 20,000 to 4,000), and three villages (populations ranging from 3,000 to 1,000). In contrast, the most asymmetric district in 1996 was Nara First ($HI_{d,t} = 0.98$), which I discussed above.

Why does this matter? The increase in district-level asymmetry that accompanied Japan's 1994 electoral reform, which redrew the boundaries of almost every single district, would have increased the amounts of money that needed to be delivered to convince voters to use their votes to compete with other municipalities. If certain LDP politicians were not delivering these larger sums, perhaps because they lacked the ability to do so, this would account for the more pronounced negative impact of $HI_{d,t}$ on support in the post-1994 period.

7.4 RECONCILING PUZZLES IN PRIOR WORK

Earlier in this chapter, we presented evidence that districts characterized by greater asymmetry in municipality size received more transfers than districts characterized by less asymmetry in municipality size. Then, we showed that despite receiving more transfers, districts characterized by greater asymmetry in municipality size exhibit systematically *lower* levels of support for the LDP, with the negative relationship between district-level asymmetry and electoral support being especially pronounced in the post-1994 era. This provides a compelling explanation for a central puzzle in Japanese politics: Why money has been found to flow to relatively *unsupportive* electoral districts, where the share of voters voting for the LDP is lower than other electoral districts. The reason this pattern of spending is puzzling is because it is at odds with the near-universal assessment of the LDP as a party that is adept at using its monopoly access to government resources to create and nurture loyal groups of supporters, who can be relied on to turn out and vote for LDP politicians, election after election.

Until now, scholars noticing this pattern have suggested that it may be attributable to a concerted effort to use government resources to buy the votes of swing voters (Horiuchi and Saito 2003). Under this

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argument, party leaders, who want to increase their party's seat share, funnel resources toward electoral districts in which the party does poorly, presumably with the intent of increasing support. However, a swing voter-based argument runs into difficulties on both theoretical and empirical grounds. Theoretically, any electoral strategy that relies on channeling benefits toward voters who do not support the party risks depleting support among all types of voters. Under this allocation rule, voters in so-called core districts would realize that feigning a lack of support for the incumbent represented their best chance of receiving more government transfers, while voters in so-called swing districts would realize that they could receive an even larger allocation if they lowered their support further. Thus, it is not difficult to see how an incumbent pursuing this strategy risks failing to capture enough votes and losing the election, which would be the worst possible outcome. Empirically, too, an electoral strategy of diverting resources to swing districts requires that party leaders have the tools to shield the transfer system from the actions of their members representing core districts, who will presumably also be agitating for more funds. There is little evidence, in either the preelectoral or postelectoral reform period, that LDP leaders wielded this kind of control.

This chapter has offered a different explanation for why money flows to districts in which LDP vote shares are lower. Importantly, this explanation does not require us to revise our understanding of the LDP as a party that funnels benefits to core supporters. In a nutshell, this pattern of resource allocations across districts is perfectly consistent with a pattern of resource allocations within districts in which government money is funneled to core supporters – not blithely, but according to a tournament in which the most supportive groups get the most money, and money declines as a function of electoral support. The reason why this allocation rule within districts leads to the opposite relationship between support and transfers across districts is because the relative sizes of the groups into which voters in a district are arranged exercises an *independent* effect on the amount of resources incumbents have to offer voters to entice them to compete against one another. In districts characterized by greater asymmetry in group size, more money is required. Thus, in these districts, incumbents offer more (money) but are also likely to settle for less (support). This produces the negative correlation between support and transfers across districts that has perplexed scholars.

For voters in Japan, the book thus far has identified two variables that impact the amount of discretionary transfers their municipalities receive. One is their municipality's position in a rank ordering of municipalities in the district constructed on the basis of electoral support. The battery

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of tests conducted in Chapter 6 show that within districts, municipalities that attain high positions in this ranking receive more transfers after the election than their counterparts attaining lower positions. The second variable is the degree of asymmetry in size of the municipalities in their district. The tests in this chapter show that municipalities in districts characterized by greater asymmetry in municipality size receive more money for a given level of support. Importantly, this could give rise to a situation in which a municipality placing first in one district receives *less* than a municipality placing last in another district. This would occur if the first municipality's district consisted of evenly sized municipalities and the latter municipality's district consisted of asymmetrically sized municipalities.

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